

HAT-P-54b

R-1160

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_231018 · created 2026-07-31T02:21:40+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460235.9449 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.048
Transit depth (Rp/Rs)^2	3.04 +/- 1.68 [%]
Orbital Inclination (inc)	89.72 +/- 1.95
Airmass coefficient 1 (a1)	0.69 +/- 0.018
Airmass coefficient 2 (a2)	0.37 +/- 0.17
Scatter in the residuals of the lightcurve fit is	2.64 %
Best Comparison Star	#2 - [552, 334]
Optimal Aperture	2.88
Optimal Annulus	7.92
Transit Duration (day)	0.0532 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.174 $\pm$ 0.048 vs 0.1572 (deviation 0.33 σ)
Duration fit vs published	0.0532 d vs 0.075 d (deviation 4.07 σ)
Mid-time O-C	-11.0 min
Depth SNR	3.5
Residual scatter	2.64 %

Light curve

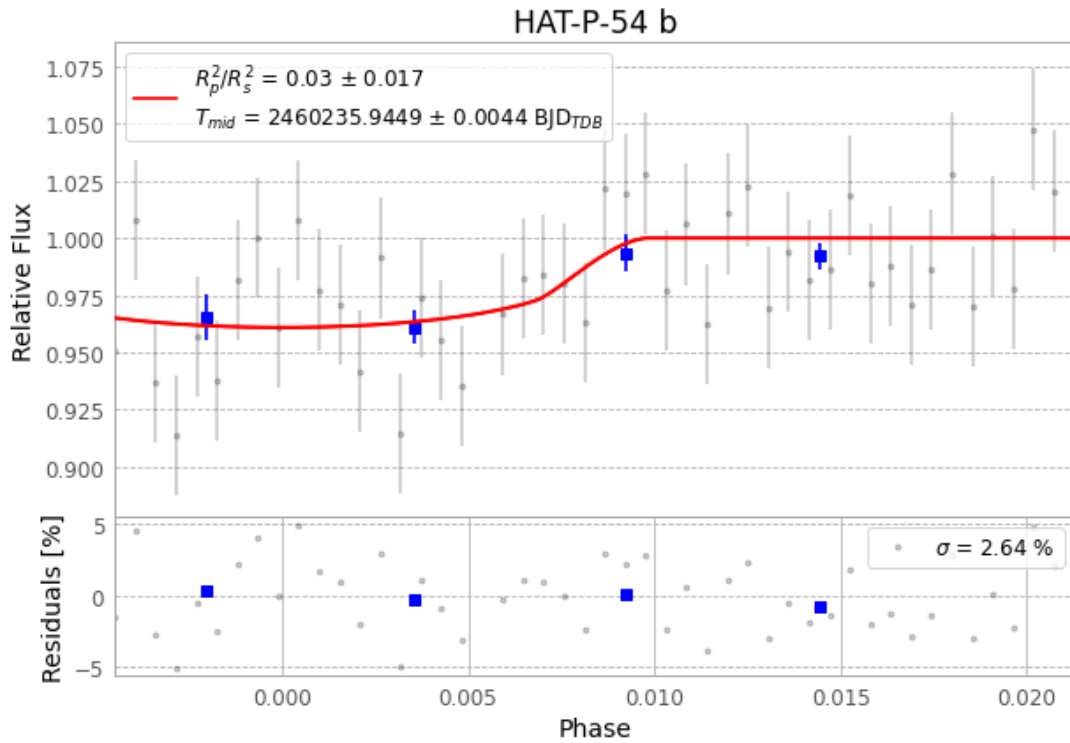


Figure 1 — FinalLightCurve_HAT-P-54 b_2023-10-18.png (SHA-256 9037dcf195337bb6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 50db94bf551b1761...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

48 FITS frames (32 MB), HATP-54231018101215.FITS → HATP-54231018123315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-231018.log (961e25b99654...), FinalLightCurve_HAT-P-54 b_2023-10-18.png (9037dcf19533...), FinalLightCurve_HAT-P-54 b_2023-10-18.csv (506401670c28...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 02:21Z.